SRTA-CITY WINTER SCHOOL (2022)

Basics of Quantum Computing and Its Applications

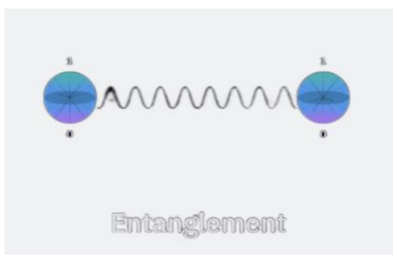
("The QC Game")

Introduction: From Planck to Shor

Earlier assumptions (classical physics):
- Light behaves as a wave.
- Electrons behave as particles (have mass).

- quantization of energy
  - Planck, ultraviolet catastrophe -- about 1900
  - Einstein, photoelectric effect -- 1905
  - Bohr's model of the atom -- 1913
- wave-particle duality (particles exhibit superposition, just like waves)
  - Schrödinger's wave equation -- 1925
  - Double Slit experiment with electrons -- 1927
- quantum physics incomplete because of non-realism, non-locality, uncertainty
  - EPR paper -- 1935
- in a specific experiment, if a certain expression C is found to be > 2, then quantum theory is true and complete
  - Bell's theorem -- 1964
  - C is always > 2
- Feynman and Mannin proposed quantum computing -- about 1980
  - only in theory, as a vague idea
- practical interest in QC
  - Deutsch --1985: QTM and first quantum algorithm
  - Shor's quantum algorithm to factor integers efficiently -- about 1994

Superposition and Entanglement



Non-realism and Uncertainty

The feature of a quantum object whereby it appears to exist in several distinct quantum states at the same time. Each object, until it is measured, will have some finite chance of being in either state.

Non-locality

It is the apparent exchange of quantum information between two particles/objects, even if the two are separated from each other by a large distance.

Toys, Mobile Games, and Visual Group Theory

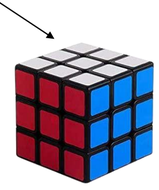
A good understanding of **quantum algorithms** requires good knowledge of **group theory** which models the underlying allowed logical operations

Toys that are popularly modeled by **Group Theory**

Dual Circles

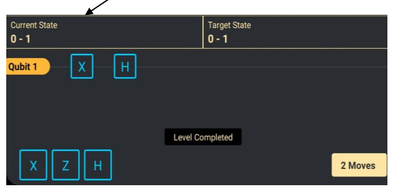
Common properties of groups:

1. **Fixed** set of movements/actions.
2. **Determinism** of actions.
3. **Reversibility**.
4. **Composability**.

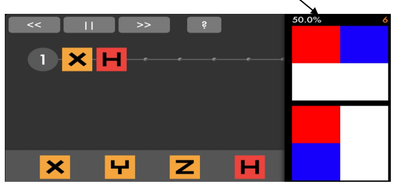


Rubik's Cube

Mobile games that make quantum computing more fun to learn.
They introduce the concepts of **qubits** and **quantum gates** to the player.

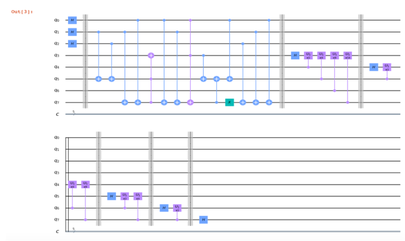


QLogic

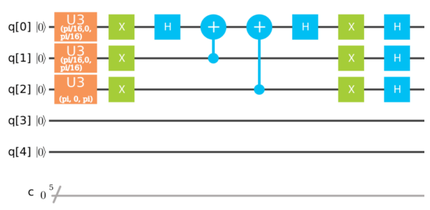


Meqanic

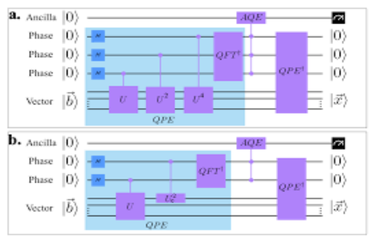
Quantum Applications: Factoring, Searching, and AI/ML



Shor's algorithm is a quantum algorithm for integer **factorization**. The time complexity of Shor's algorithm to decompose integer N on a quantum computer is O($log^2 N$).



Grover's algorithm, also known as the **quantum search** algorithm, refers to a quantum algorithm for unstructured search. Grover's algorithm takes O(√N) time.



The **HHL algorithm** solves a system of linear equations and is used as a component in the machine learning classification algorithm SVM to define a faster quantum version of the algorithm called QSVM. HHL depends on the availability of QRAM.

Hands-on Quantum Computing

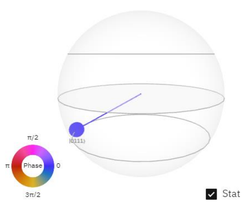
Real Quantum Computers

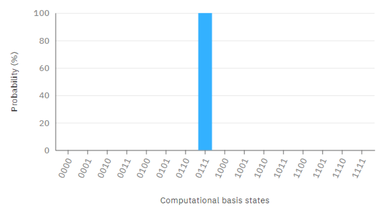
E.g., IBM Q Experience

```
OPENQASM 2.0;
include "qelib1.inc";

qreg q[4];
creg c[4];

x q[0];
x q[1];
x q[3];
ccx q[0],q[1],q[2];
ccx q[0],q[1],q[3];
ccx q[3],q[2],q[1];
```



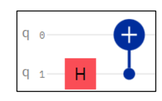


QC Simulators (Reproducible Experiments)

Programmatic
E.g., QC Engine

```
qc.reset(2);
qc.write(0b00);
qc.had(0b10);
qc.cnot(0b01, 0b10);
var result = qc.read();
```

Graphical
E.g., IBM Quantum Composer



Bibliography

1. **Programming Quantum Computers, Eric R. Johnston, Nic Harrigan, Mercedes Gimeno-Segovia, 2019.**
2. **Explorations in Quantum Computation, 2nd Edition, Colin P. Williams, 2011.**
3. **Visual Group Theory, Nathan Carter, 2009.**
4. **The Princeton Companion to Mathematics, Tim Gowers (Ed.), 2008.**

Team

- **Team Lead:**
  Dr. Moez A. AbdelGawad
  moez@alexu.edu.eg
- **Jehad Aly (Fac. of Eng.)**
- **Nada Rabea (Fac. of Sci.)**
- **Noha Ahmed (Fac. of Eng.)**
- **Reem Ahmed (Fac. of Sci.)**